

DATA FLOW, QUALITY AND STANDARDS

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ABOUT ME



- **Nature Conservation Agency of the Czech Republic (AOPK ČR)** – since 2020
- **Biodiversa+** – lead development and coordination of the **BiodivPond** pond-monitoring pilot
- **TAIEX** – expert mission to Kosovo, December 2023
- **ConNatur LIFE** – with Ukrainian partners, the path to Natura 2000 and to implementing the Birds and Habitats Directives in Ukraine

A NATIONAL DATABASE IN FIVE LAYERS

Reference layer – every other layer points here

- national **checklist**, with **crosswalks** to the EU and GBIF taxonomic backbones
- **protection and Red List status**, by taxon ID
- **grids** and **protected sites**
- **habitat crosswalk** and **vocabularies**

1 • Staging – raw data exactly as received. Never edited.



2 • Validation – automatic checks and expert review. A status on every record.

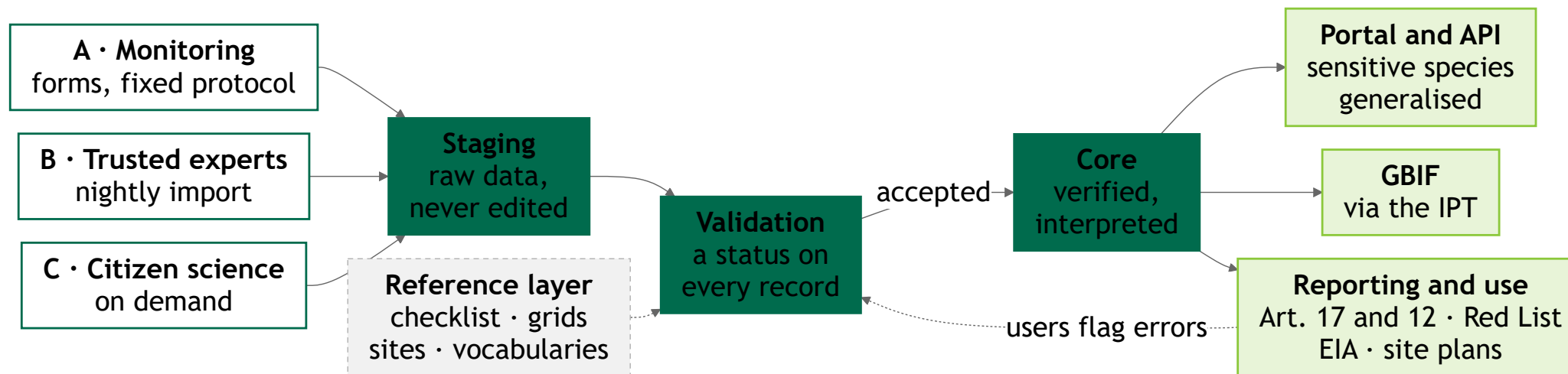


3 • Core – verified, interpreted records. The one copy everybody uses.



4 • Publication – portal with sensitive species generalised • API • GBIF via the IPT • reporting

THREE SOURCES, ONE PATH



The layers are the same for every source. What differs is how much checking a record needs – and who does it.

FLOW A – STRUCTURED MONITORING

- **Checks built into the form:** required fields, pick-lists fed from the checklist and vocabularies, device position with its accuracy
- **Automatic sync** to staging – nobody retypes
- **The coordinator validates:** protocol followed, all visits done, values plausible
- **Stored as an event:** one visit, with its occurrences and measurements

One visit, stored as an event

Event	Visit · pond 07 · 14 May · night torch count · 2 observers, 2 h
Occurrence	<i>Bombina variegata</i> – present, 12 adults
Occurrence	crested newt (<i>Triturus</i> sp.) – absent
Measurement	water depth – 0.6 m
Measurement	shading – 30 %

Illustrative values. The event carries the date, site, method and effort; the absence is only meaningful because of them.

FLOW B – RECORDS FROM TRUSTED EXPERTS

- **Trusted status is earned:** accreditation per species group, reviewed
- **Automatic overnight import** from their app or database
- **Automatic checks:** name matches the checklist · point inside the country, plausible uncertainty · plausible date and season · far outside the known range?
- **Clean → accepted. Flagged → review queue**, not rejected

A status on every record: the NDOP example

Validation – proposed by a regional expert	
0	not yet validated
1	proposed as fully reliable
3	proposed as less reliable
6	proposed as erroneous
9	proposed for correction

A central guarantor then confirms the proposal. For species protected by Czech law, species of Community interest and Annex I birds, this check is compulsory.

FLOW C – CITIZEN SCIENCE, ON DEMAND

- **Imported when needed** – for one assessment, taxon or area
- **Filters first:** open licence · community identification grade · photo
- **Expert validation** before use in any official product
- **Source always visible:** `datasetName`, `basisOfRecord`, `identificationVerificationStatus`
- **Deduplicate** – one sighting often arrives by two routes

What stays on the record after import

<code>datasetName</code>	iNaturalist Research-grade Observations
<code>basisOfRecord</code>	HumanObservation
<code>identificationVerificationStatus</code>	community consensus – expert check pending
<code>license</code>	CC BY-NC 4.0
<code>associatedMedia</code>	link to the photograph
<code>occurrenceID</code>	the platform’s own ID, kept unchanged

Illustrative. `identificationVerificationStatus` takes free text in Darwin Core, so agree a short national list of values for it.

DATA THAT NOBODY USES DO NOT GET CORRECTED

Build the way back in

- **Any user can flag** a record
- **The validator** for that group reviews it
- **The observer is told** – and often knows the answer
- **Every change is versioned:** who, when, why – an audit trail

The uses that find the errors

- **Art. 17 / Art. 12 reporting** – with EU accession
- **Natura 2000 / Emerald** site assessments
- **Red List** – AOO and EOO from the records (R package *ndopred*)
- **EIA and permitting**
- **Protected-area management plans**



SPEAK DARWIN CORE FROM DAY ONE

- **Use DwC directly** – or keep a documented **crosswalk** from every internal field
- **Occurrence** and **Event** cores; **Extended Measurement or Fact; Humboldt** for effort and completeness
- **Fixed values:** `basisOfRecord`; `occurrenceStatus` = present / absent
- **Mandatory:** `coordinateUncertaintyInMeters`; persistent `occurrenceID` and `eventID`
- **Pays back:** GBIF without rework · EU reporting and INSPIRE · open tools · no vendor lock-in · cross-border exchange

A crosswalk: survey spreadsheet → Darwin Core

Your column	Darwin Core term
Species	<code>scientificName</code> + <code>taxonID</code>
Date	<code>eventDate</code> (2026-06-18)
Lat / Long	<code>decimalLatitude</code> / <code>decimalLongitude</code>
GPS accuracy (m)	<code>coordinateUncertaintyInMeters</code>
Found? yes / no	<code>occurrenceStatus</code> : present / absent
Record no.	<code>occurrenceID</code> – never reused

The full crosswalk, with the event and measurement fields, is on the handout.

GIS STRUCTURE – ONE GEOMETRY, MANY RECORDS

Oracle Spatial (SDO) × ArcGIS geodatabase (SDE)

- **Geometry stored as SDO_GEOMETRY** – ArcGIS, QGIS, GDAL and plain SQL all read it. Esri's default, *ST_Geometry*, needs Esri's libraries for SQL
- **Tables registered with the geodatabase** – ArcGIS edits the table SQL reads; no copies to keep in sync
- **One geometry type per table** – points, lines, polygons apart; ArcGIS takes no mix
- **Same coordinate system, tolerance and spatial index** on both sides
- **Versioned edits are invisible to SQL** until posted – version only what must be

1 geometry : N records

Geometry	G-1042 · polygon · pond 07 with its shore
Occurrence	<i>Bombina variegata</i> – 14 May 2026
Occurrence	<i>Bombina variegata</i> – 3 June 2027

Geometry	H-0215 · polygon · meadow segment
Habitat	6510 lowland hay meadows – 70 %
Habitat	6430 tall-herb fringes – 30 %

Illustrative. A record holds its geometry's ID, never a shape of its own: drawn once, corrected once, and every record on it follows.

SIX HABITS OF A DATABASE THAT LASTS

1

A named data steward – and validators for each species group

Here: **to be named**

2

Funded surveys deliver DwC-ready data – written into every contract

3

Raw data never overwritten – provenance on every record

4

A written sensitive-species policy – generalise, do not hide

5

Clear licence and citation – so contributors get credit

6

Start simple, then iterate – one flow, one group, one year

FOUR THINGS TO TAKE AWAY

- 1 Layers protect you** – keep the raw data, validate, then publish.
- 2 Every record carries a status and a source.**
- 3 Use is the best quality control** – build the way back in.
- 4 Darwin Core from day one** – every field mapped.

For discussion – in Kosovo*: which flow comes first? **answer from the room** · Who owns the national checklist? **to be agreed** · Who validates each species group? **to be agreed**

*This designation is without prejudice to positions on status, and is in line with UNSCR 1244/1999 and the ICJ Opinion on the Kosovo declaration of independence.

THANK YOU FOR YOUR ATTENTION!

github.com/jonasgaigr/kosovo-biodiversity-data-workshop-2026



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